

AI-driven prediction of heat transfer coefficients for GT blade cooling

Semester Project at the Laboratory for Applied Mechanical Design (LAMD)

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Context:



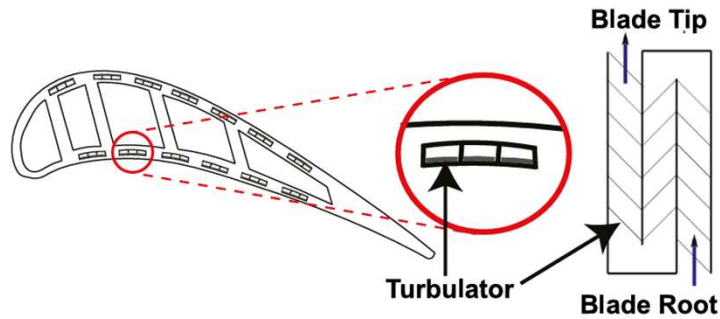
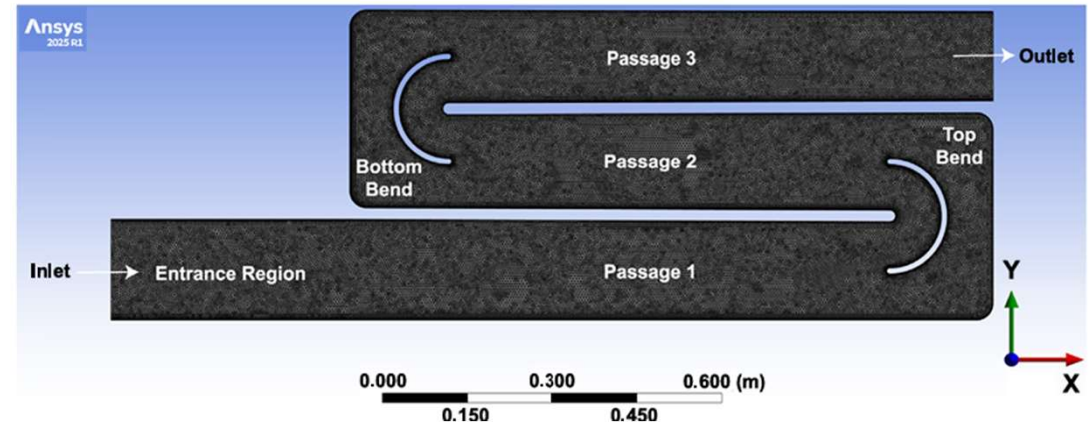
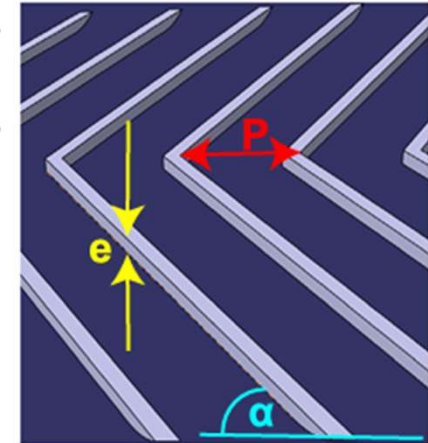


Fig. 1. Conceptual illustration of wall-integrated cooling passages in a turbine blade, reproduced from [3]. Left: top view of a blade showing cooling passages embedded within the blade wall. Center: enlarged top view of an individual channel highlighting the rib turbulators. Right: schematic side view of the ribbed cooling channel along the blade span (root to tip). The drawings are schematic and not to scale.

- Internal cooling of GT blades via **rib-turbulated channels**
- Ribs **enhance heat transfer** at the cost of **increased pressure drop**
- Design space parametrized by **5 variables: AR, α , e/D_h , P/e , Re**
- **50 CFD cases** sampled via Latin hypercube for good design space coverage



Parameter	V-ribs
Aspect ratio (AR)	4–15
Guide vane span ($^\circ$)	180, 135
Rib angle (α)	30–75
e/D_h (%)	5–20
P/e	5–15
Reynolds number ($\times 10^3$)	20–200
Number of cases	50 G180°



Objectives:

- Minimize Pressure Drop

$$\Delta P = P_{\text{in}} - P_{\text{out}}$$

- Maximize Heat Transfer Coefficient (HTC)

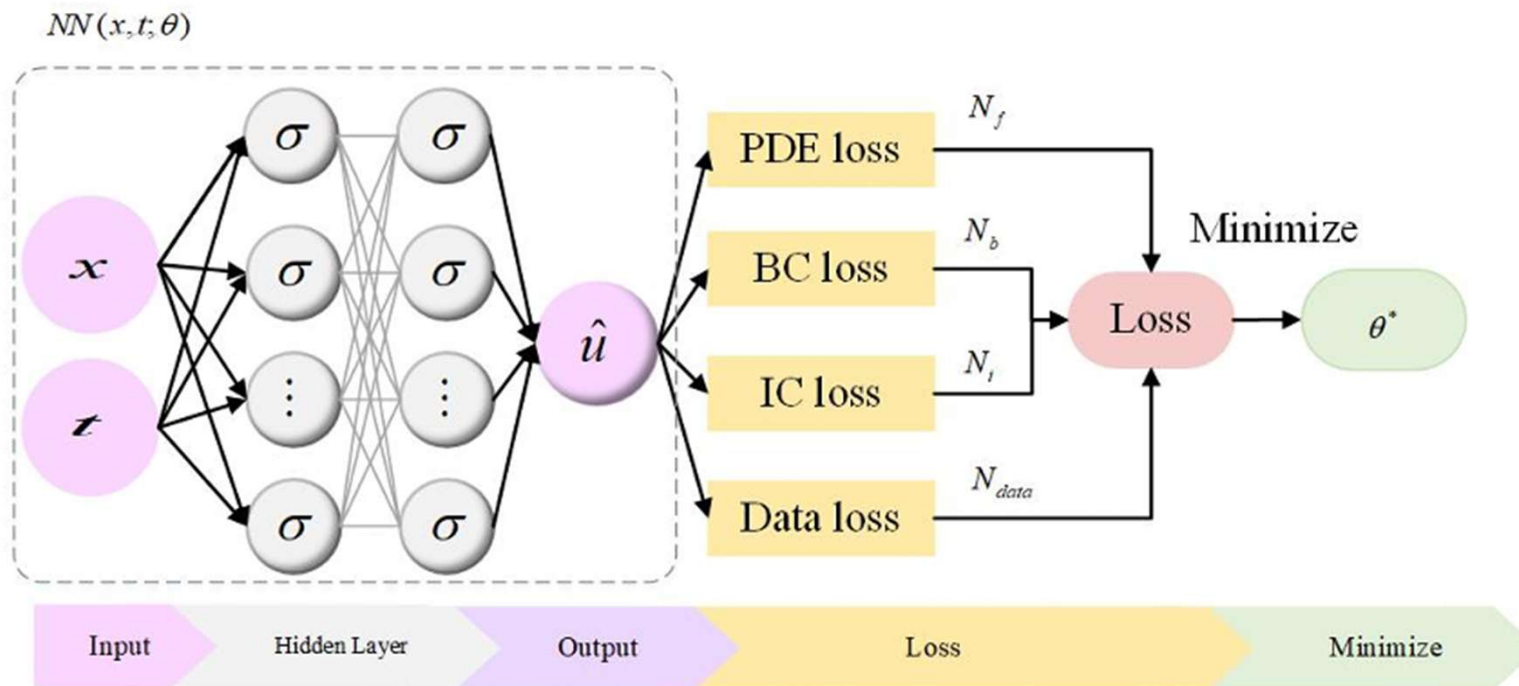
$$\text{HTC} = \frac{q''}{T_w - T_b}$$

From the PINN

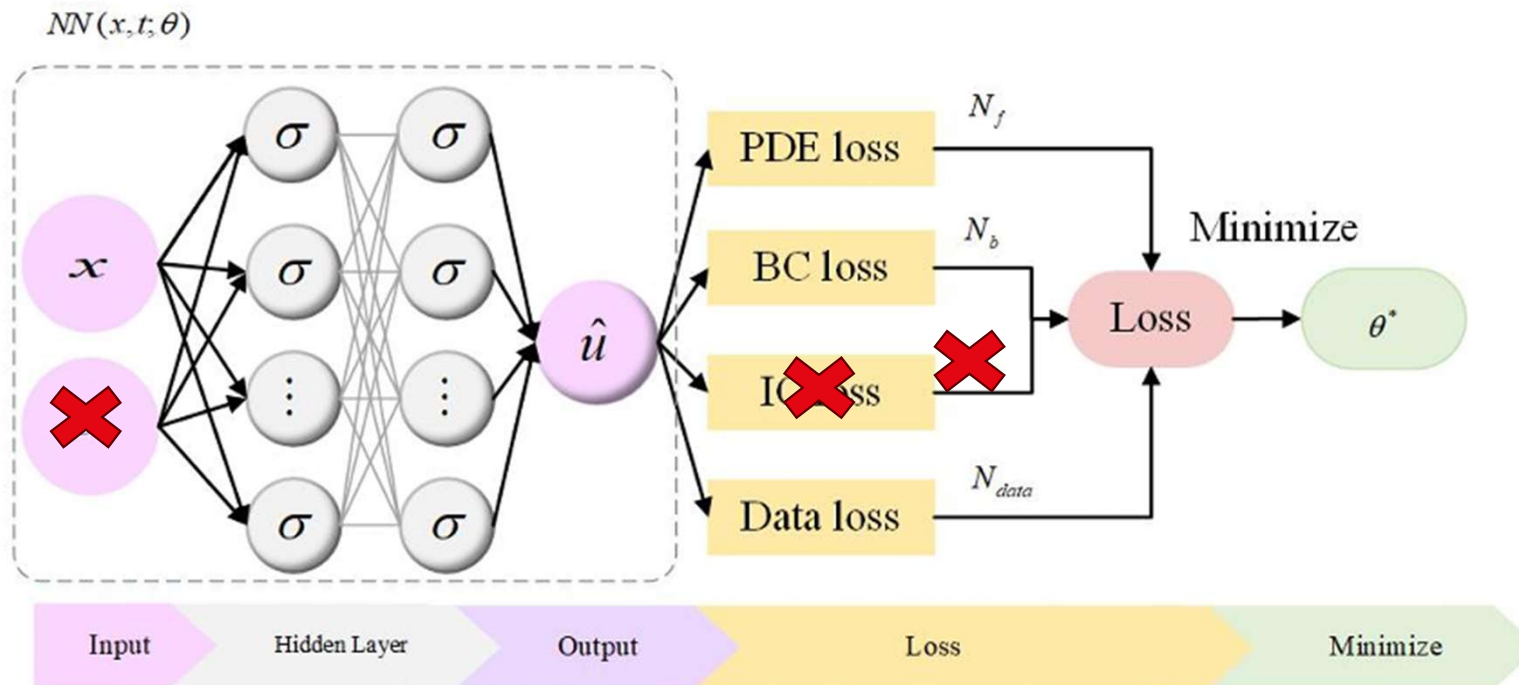
From the MLP

I. PINN to predict all fields:





PINN, steady state, 3D, Heat equation



PINN, steady state, 3D, Heat equation

- **PDE:** continuity, Navier-Stokes, energy equation
- **BC:** inlet velocity profile, inlet temperature, wall temperatures, outlet pressure
- **Supervised:** direct fit to CFD data for all fields at a sparse set of points (500 over 10 000 000 points per DP)

$$\begin{aligned}
 \mathcal{L} = & w_{\text{pde}} \underbrace{(\mathcal{L}_{\text{div}} + \sum_i \mathcal{L}_{\text{mom},i} + \mathcal{L}_{\text{heat}})}_{\mathcal{L}_{\text{pde}}} \\
 & + w_{\text{bc}} \underbrace{(\sum_i \mathcal{L}_{\text{in},u_i} + \mathcal{L}_{\text{in},T} + \mathcal{L}_{\text{out},p} + \sum_i \mathcal{L}_{\text{wall},u_i} + \mathcal{L}_{\text{wall},T})}_{\mathcal{L}_{\text{bc}}} \\
 & + w_{\text{sup}} \underbrace{(\sum_i \mathcal{L}_{\text{sup},u_i} + \mathcal{L}_{\text{sup},T})}_{\mathcal{L}_{\text{sup}}} + w_{\text{sup},p} \mathcal{L}_{\text{sup},p}
 \end{aligned}$$

Table 1: PINN loss terms. $\overline{(\cdot)}_{\Omega}$ is the mean (MSE) over points in Ω . σ_{ϕ}^2 is the field variance from training data. $\tilde{\mu} = \mu(\tilde{T})/\mu_{\text{ref}}$ (Sutherland).

Category	Name	Notation	Formula
PDE	Continuity	$\mathcal{L}_{\text{div}}$	$\overline{(\nabla \cdot \tilde{\mathbf{u}})}_{\Omega_c}^2$
	Navier-Stokes ($i=x, y, z$)	$\mathcal{L}_{\text{mom},i}$	$\overline{\left(\tilde{\mathbf{u}} \cdot \nabla \tilde{u}_i + \partial_{x_i} \tilde{p} - \frac{\nu \tilde{\mu}}{V_{\text{in}}} \nabla^2 \tilde{u}_i \right)}_{\Omega_c}^2$
	Energy	$\mathcal{L}_{\text{heat}}$	$\overline{\left(\tilde{\mathbf{u}} \cdot \nabla \tilde{T} - \frac{\alpha}{V_{\text{in}}} \nabla^2 \tilde{T} \right)}_{\Omega_c}^2$
BC	Inlet velocity	$\mathcal{L}_{\text{in},u_i}$	$\frac{1}{\sigma_{\tilde{u}_i}^2} \overline{(\tilde{u}_i - \tilde{u}_i^{\text{CFD}})}_{\Gamma_{\text{in}}}^2$
	Inlet temperature	$\mathcal{L}_{\text{in},T}$	$\frac{1}{\sigma_{\tilde{T}}^2} \overline{(\tilde{T} - 1)}_{\Gamma_{\text{in}}}^2$
	Outlet pressure	$\mathcal{L}_{\text{out},p}$	$\frac{1}{\sigma_{\tilde{p}}^2} \overline{\tilde{p}}_{\Gamma_{\text{out}}}^2$
	Wall no-slip	$\mathcal{L}_{\text{wall},u_i}$	$\frac{1}{\sigma_{\tilde{u}_i}^2} \overline{\tilde{u}_i^2}_{\Gamma_w}$
	Wall temperature	$\mathcal{L}_{\text{wall},T}$	$\frac{1}{\sigma_{\tilde{T}}^2} \overline{\tilde{T}^2}_{\Gamma_w}$
Supervised	Velocity	$\mathcal{L}_{\text{sup},u_i}$	$\frac{1}{\sigma_{\tilde{u}_i}^2} \overline{(\tilde{u}_i - \tilde{u}_i^{\text{CFD}})}_{\Omega_{\text{sup}}}^2$
	Pressure	$\mathcal{L}_{\text{sup},p}$	$\frac{1}{\sigma_{\tilde{p}}^2} \overline{(\tilde{p} - \tilde{p}^{\text{CFD}})}_{\Omega_{\text{sup}}}^2$
	Temperature	$\mathcal{L}_{\text{sup},T}$	$\frac{1}{\sigma_{\tilde{T}}^2} \overline{(\tilde{T} - \tilde{T}^{\text{CFD}})}_{\Omega_{\text{sup}}}^2$

PINN, steady state, 3D, Heat equation

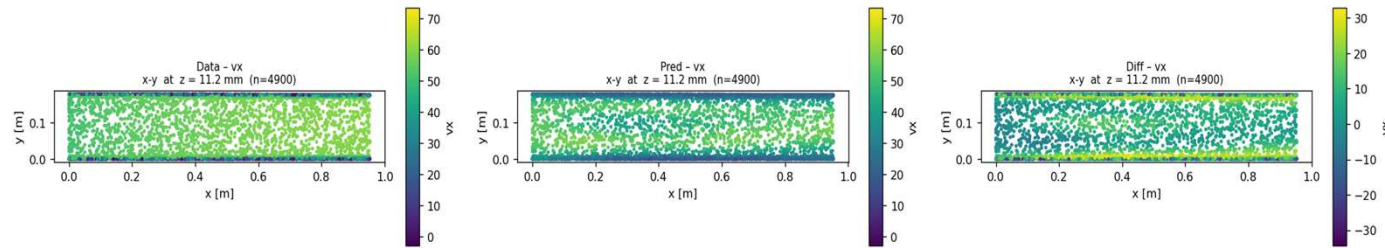
Manual weighting	Hand-tuned scalars w_{pde} , w_{bc} , w_{sup}	Tedious and case specific: doesn't transfer across geometries or Re
Learnable weights	Weights as learnable parameters with min-max bounds $L_{\text{tot}} = w_{\text{pde}}L_{\text{pde}} + w_{\text{bc}}L_{\text{bc}} + w_{\text{sup}}L_{\text{sup}}$	Collapsed toward bounds: no meaningful gain over fixed weights
SoftAdapt	Dynamic rescaling by rate of change across iterations	Marginal improvement, added hyperparameter sensitivity
Conclusion	No specific weighting: focus on scaling from the start	More robust: all loss terms naturally $O(1)$ via non-dimensionalization and z-score standardization

EPFL PINN V1: Unidirectional, simple isothermal channel

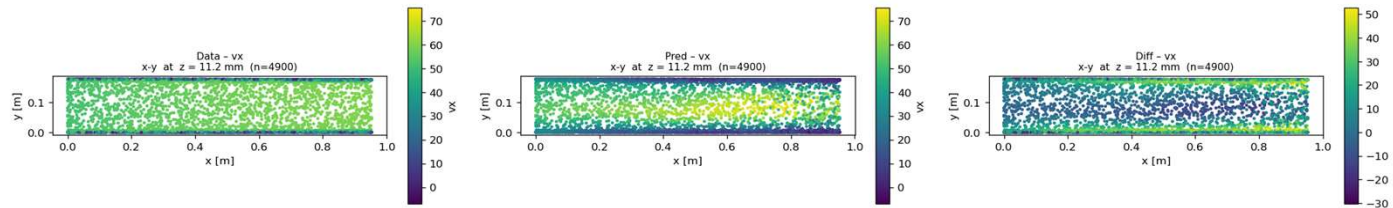
10

Velocity field (V_x): CFD data, PINN prediction, error

With data:

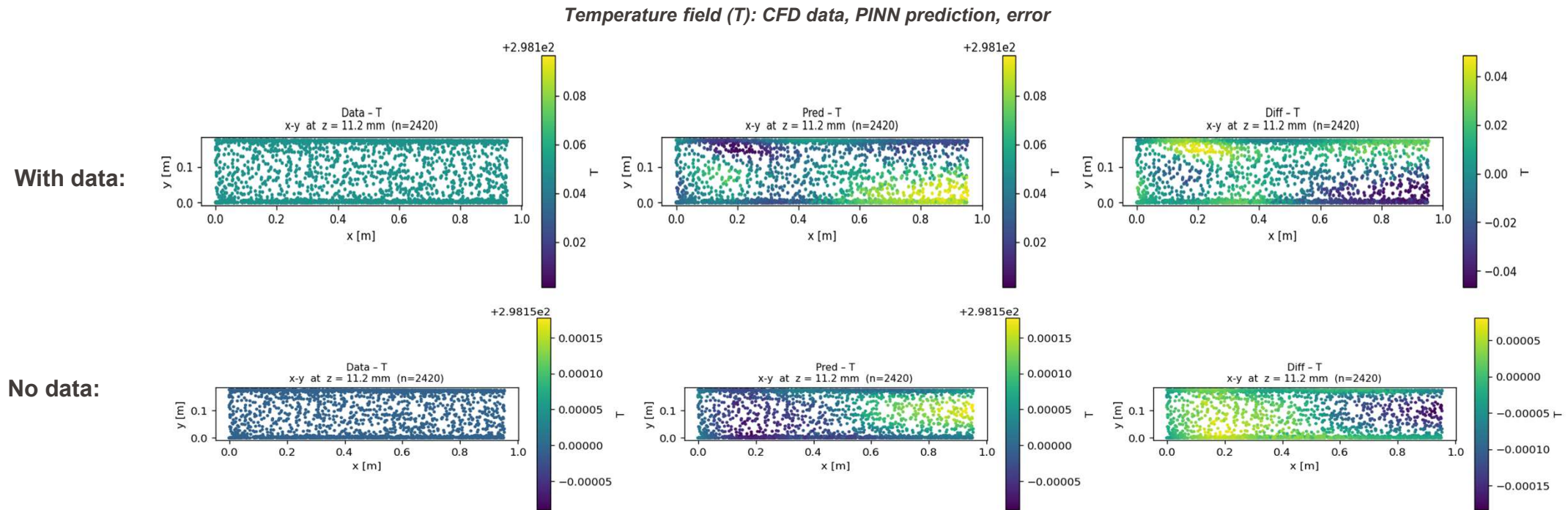


No data:



$$\begin{pmatrix} X \\ y \\ Z \end{pmatrix} \xrightarrow{\text{PINN}} \begin{pmatrix} V_x \\ V_y \\ V_z \\ P \\ T \end{pmatrix}$$

- Velocity field well recovered in both cases
- Near-zero error inside the channel
- Higher errors concentrated near the boundary layer: expected, since the **no-slip BC** is enforced exactly by the PINN while CFD wall-adjacent cells don't reach zero velocity



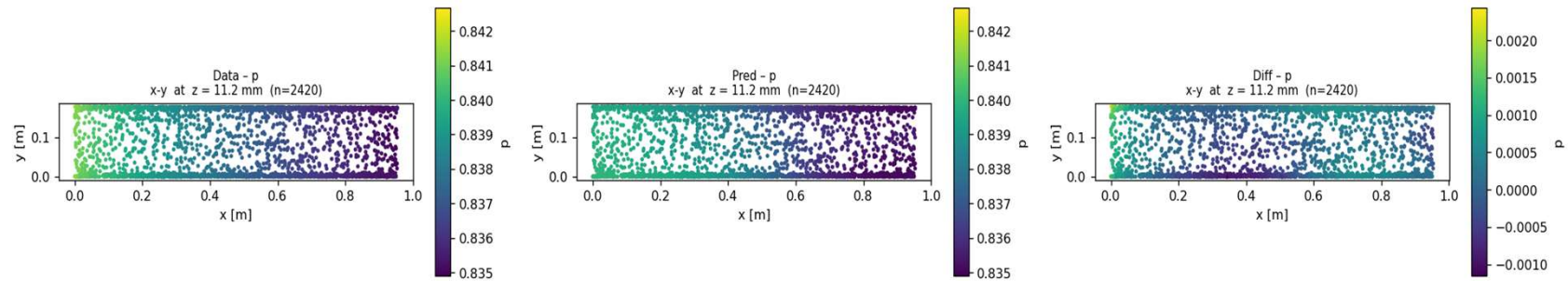
- **Isothermal channel**
- With supervised data: **errors $\sim 0.01^\circ$** , without: even smaller (redundancy physics losses VS supervised losses)
- Color maps can be misleading: **absolute errors are negligible**

EPFL PINN V1: Unidirectional, simple isothermal channel

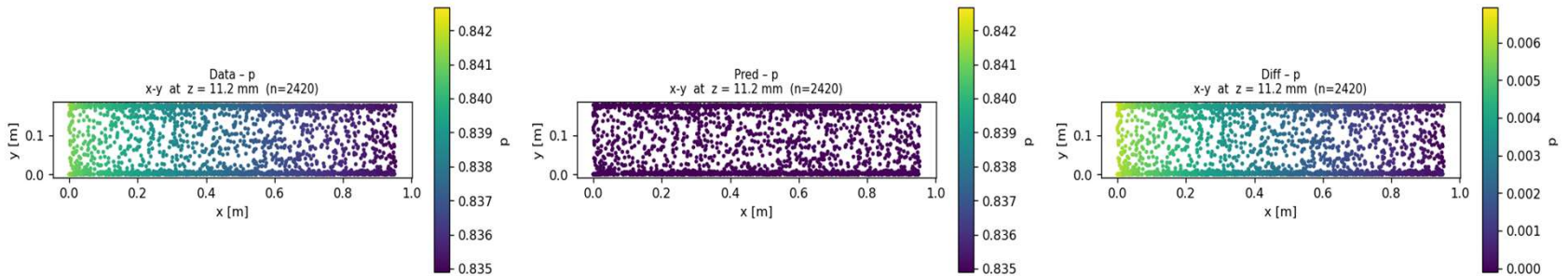
12

Pressure field (P): CFD data, PINN prediction, error

With data:



No data:



$$\frac{D\mathbf{u}}{Dt} = \nu \nabla^2 \mathbf{u} - \nabla \frac{p}{\rho} + \frac{1}{\rho} \mathbf{f},$$

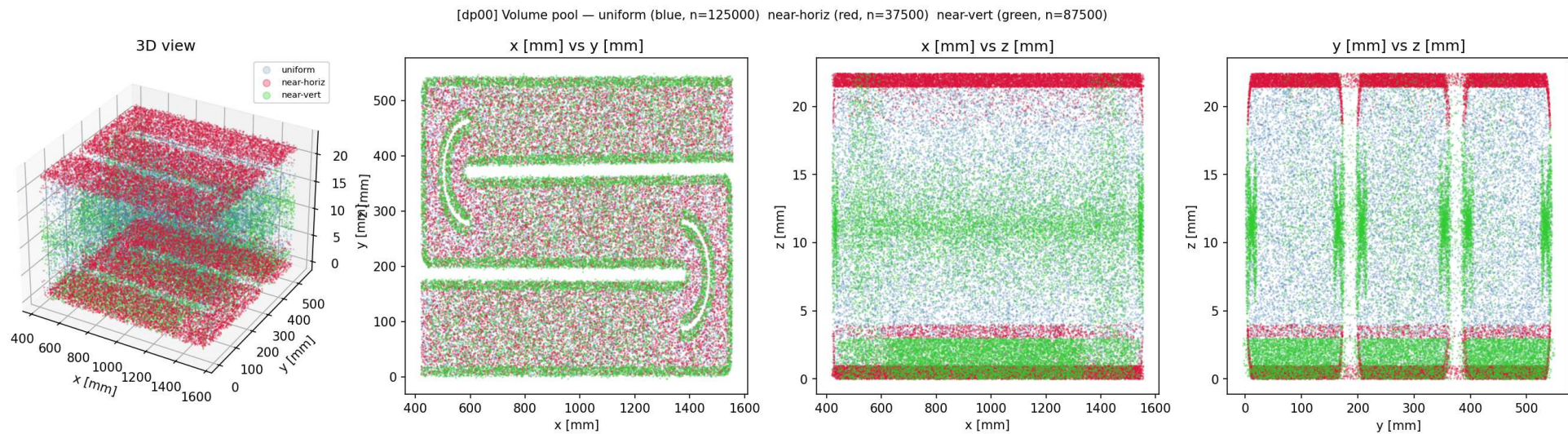
- With supervised data: pressure well recovered, **errors ~0.001 Pa**
- **Without data:** model collapses to a **constant pressure field**

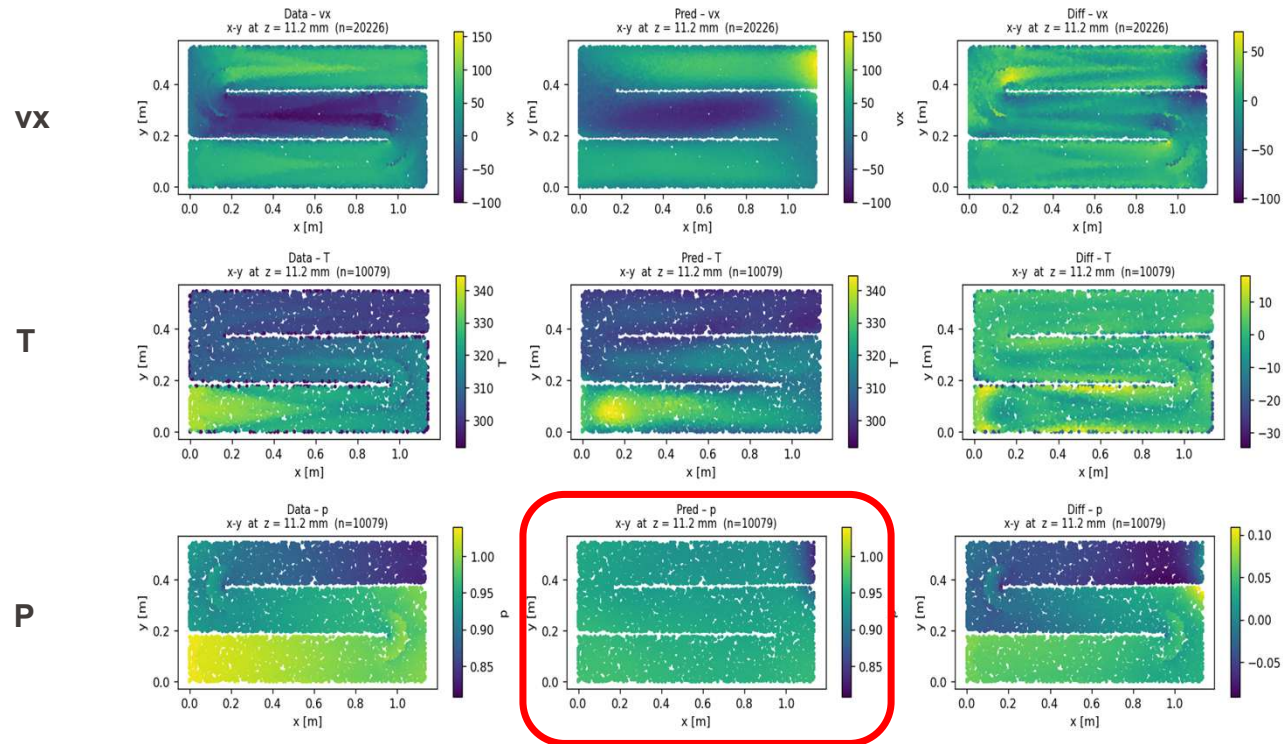
➔ **Supervised data is essential** to select the physically correct solution, not just a mathematically valid one

PINN V2: G180deg guide-vane configuration

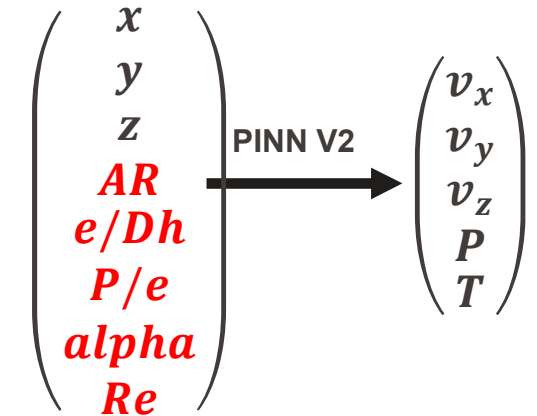
- **Naive sampling:** BCs not respected
- **Adaptive sampling:** 50% of collocation points are close to the walls (within 1mm of the horizontal walls, and within 20mm of the vertical walls, proportional to channel aspect ratio ($L_x \gg L_y$))
- **5 parameters** added as inputs $\rightarrow$ single PINN trained on all 50 design points simultaneously
- Results shown are inferences on **4 unseen design points**

$$\begin{pmatrix} x \\ y \\ z \\ \text{AR} \\ e/Dh \\ P/e \\ \alpha \\ Re \end{pmatrix} \xrightarrow{\text{PINN V2}} \begin{pmatrix} v_x \\ v_y \\ v_z \\ P \\ T \end{pmatrix}$$

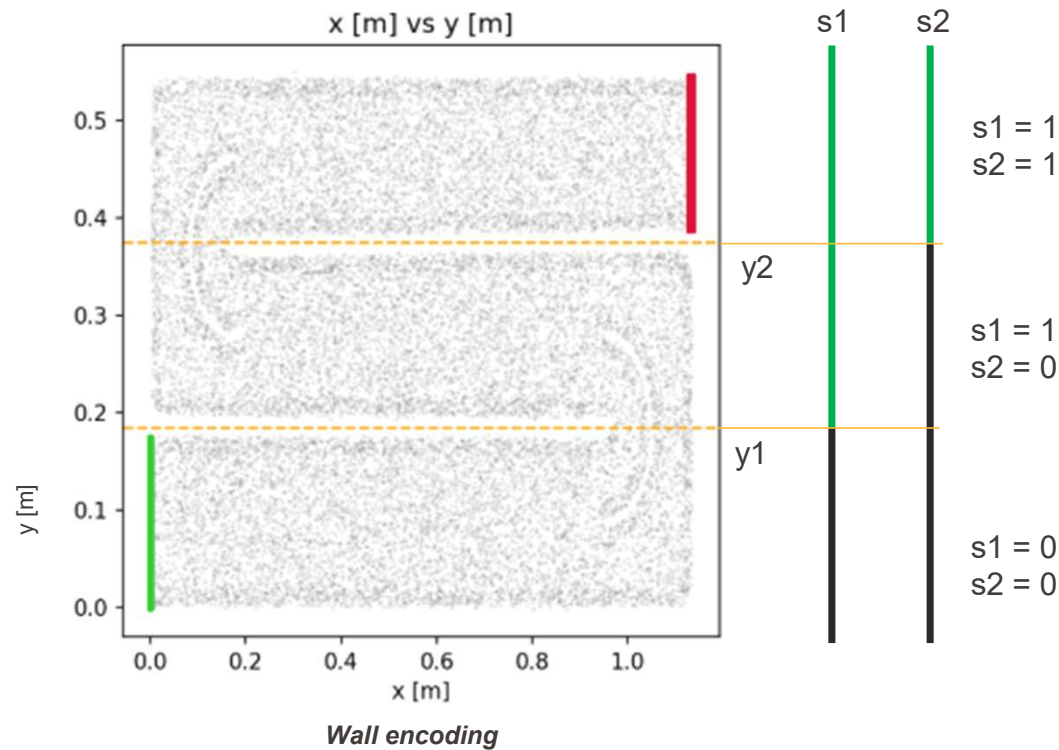




V_x , T , P : CFD, PINN prediction, error



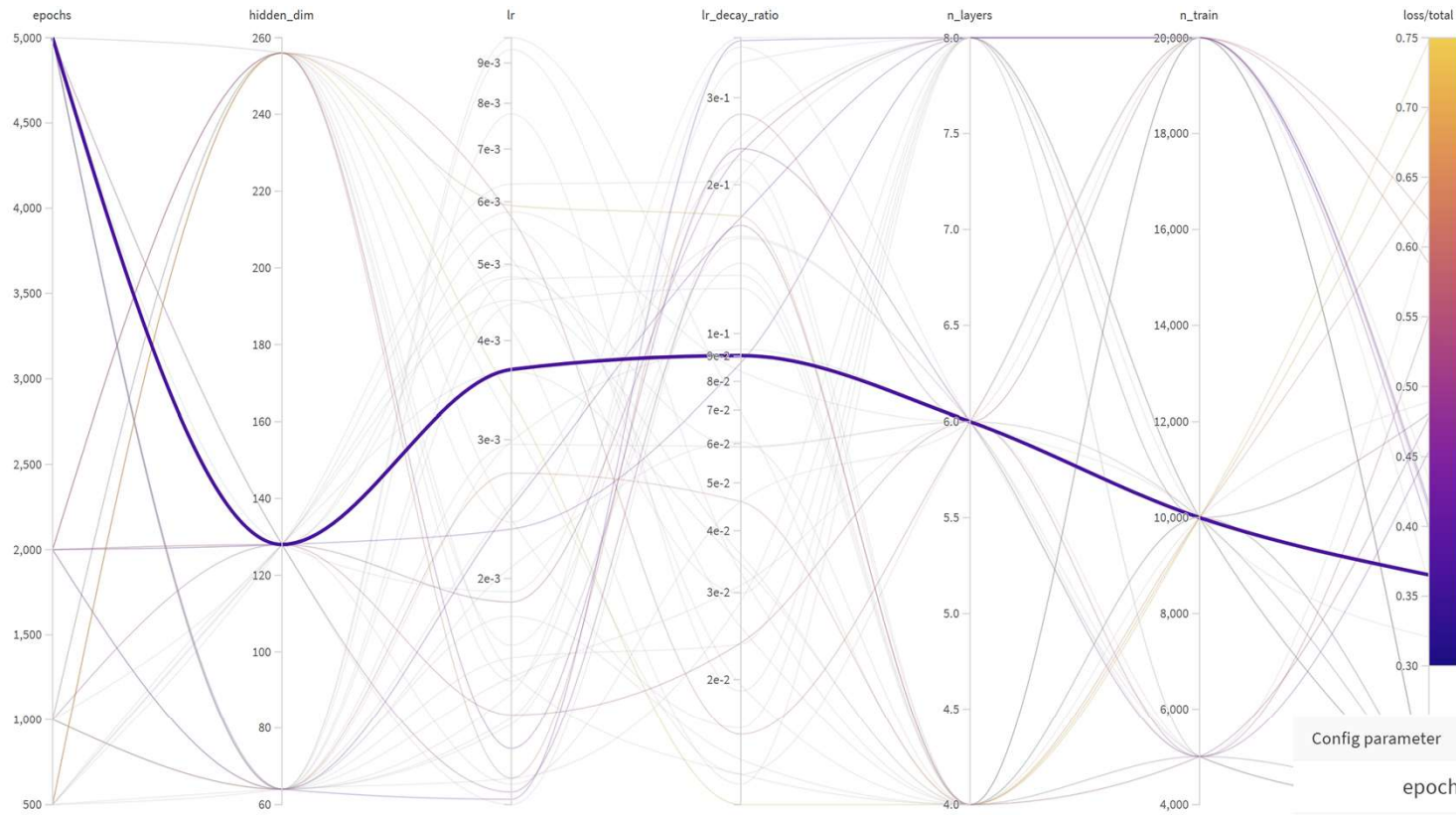
- Velocity and temperature: well recovered
- Pressure is wrong: predicted field goes straight through the walls
- Root cause: velocity and temperature are strongly constrained (NS equation, heat equation, inlet, wall BCs), pressure only has the outlet BC and its gradient in NS



$$\begin{pmatrix} x \\ y \\ z \\ AR \\ e/Dh \\ P/e \\ \alpha \\ Re \\ \textcolor{red}{s1} \\ \textcolor{red}{s2} \end{pmatrix} \xrightarrow{\text{PINN V3}} \begin{pmatrix} v_x \\ v_y \\ v_z \\ P \\ T \end{pmatrix}$$

$$s1 = \sigma\left(\frac{y - y_{wall1}}{\epsilon}\right) \quad s2 = \sigma\left(\frac{y - y_{wall2}}{\epsilon}\right)$$

- **Problem:** PINN has no direct information about wall geometry
- **Solution:** two extra input features $s1$, $s2$: steep sigmoids centered at each wall position ($\epsilon = 0.002$ m)
 → Act as **near-threshold functions**: switch $0 \rightarrow 1$ at each wall, encoding which pass the point belongs to



Hyperparameter search: parallel coordinates plot and correlation table

Chosen hyperparameters:

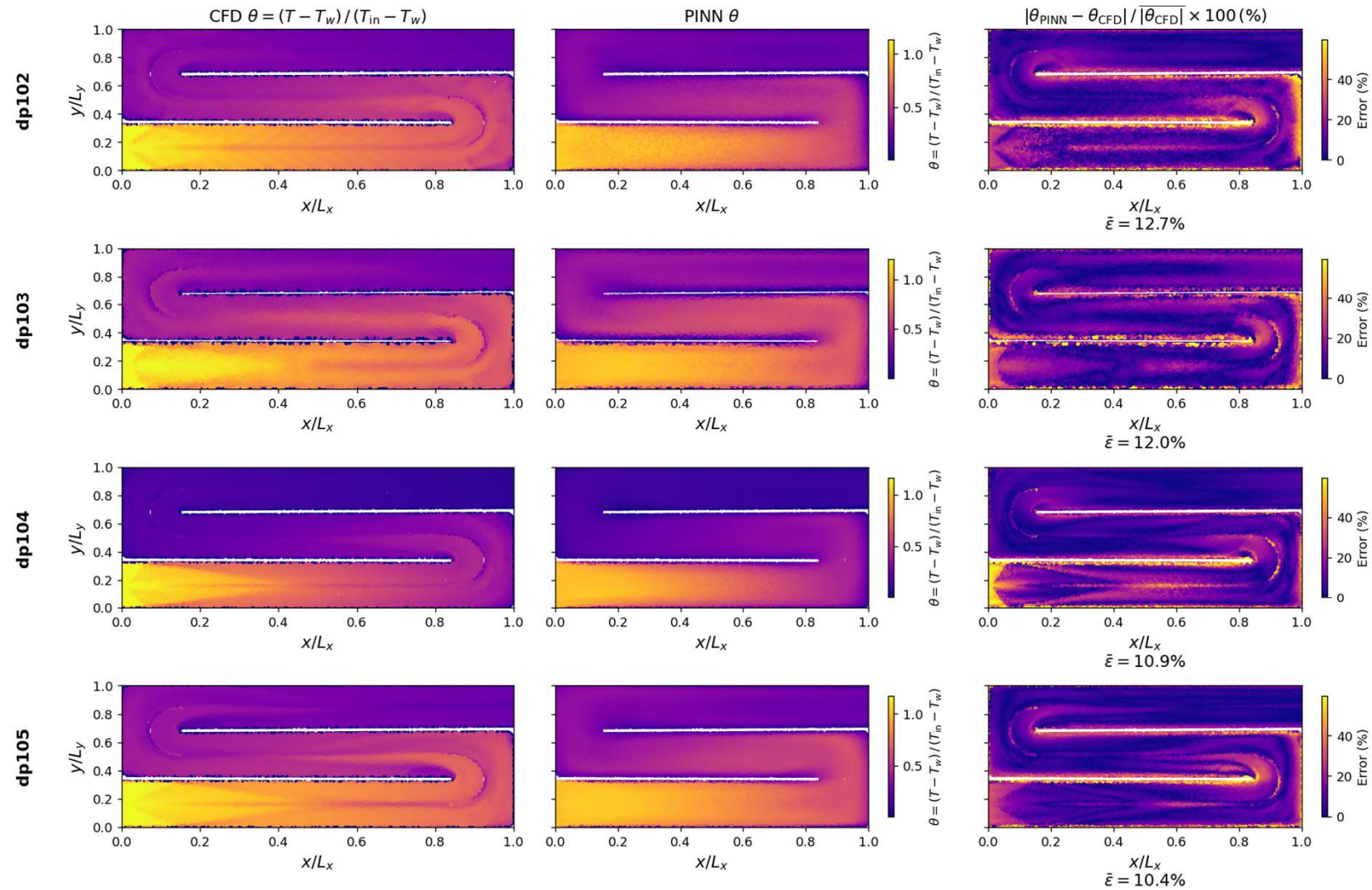
- epochs: 5,000
- hidden_dim: 128
- lr: 0.0036
- lr_decay_ratio: 0.09
- n_layers: 6
- n_train: 10,000

→ Training in 10 hours

→ Inference in 20 seconds!

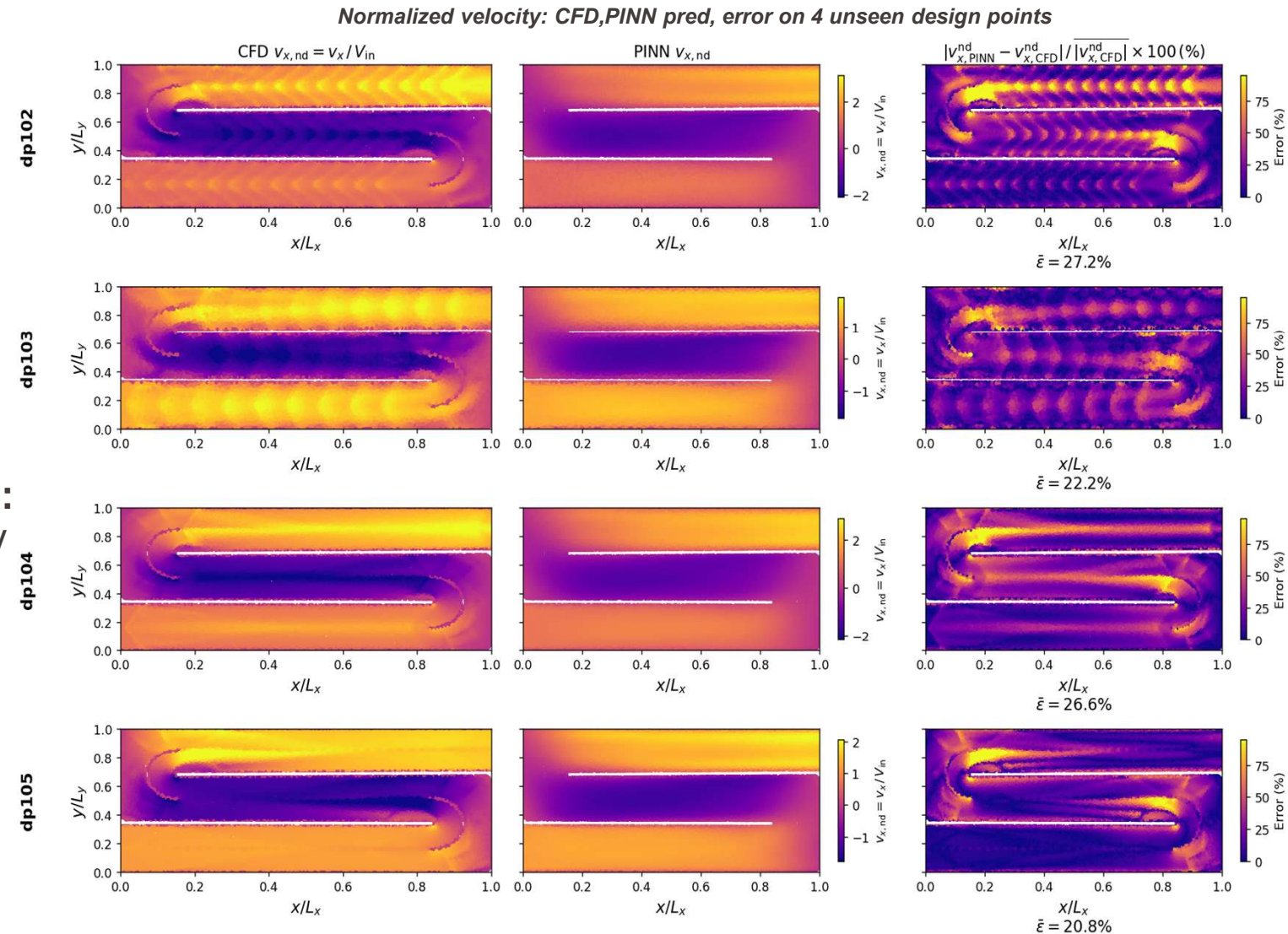
Config parameter	Importance ⓘ ↓	Correlation
epochs		
n_layers		
n_train		
lr_decay_ratio		
lr		
hidden_dim		

Normalized Temperature (θ): CFD, PINN pred, error on 4 unseen design points

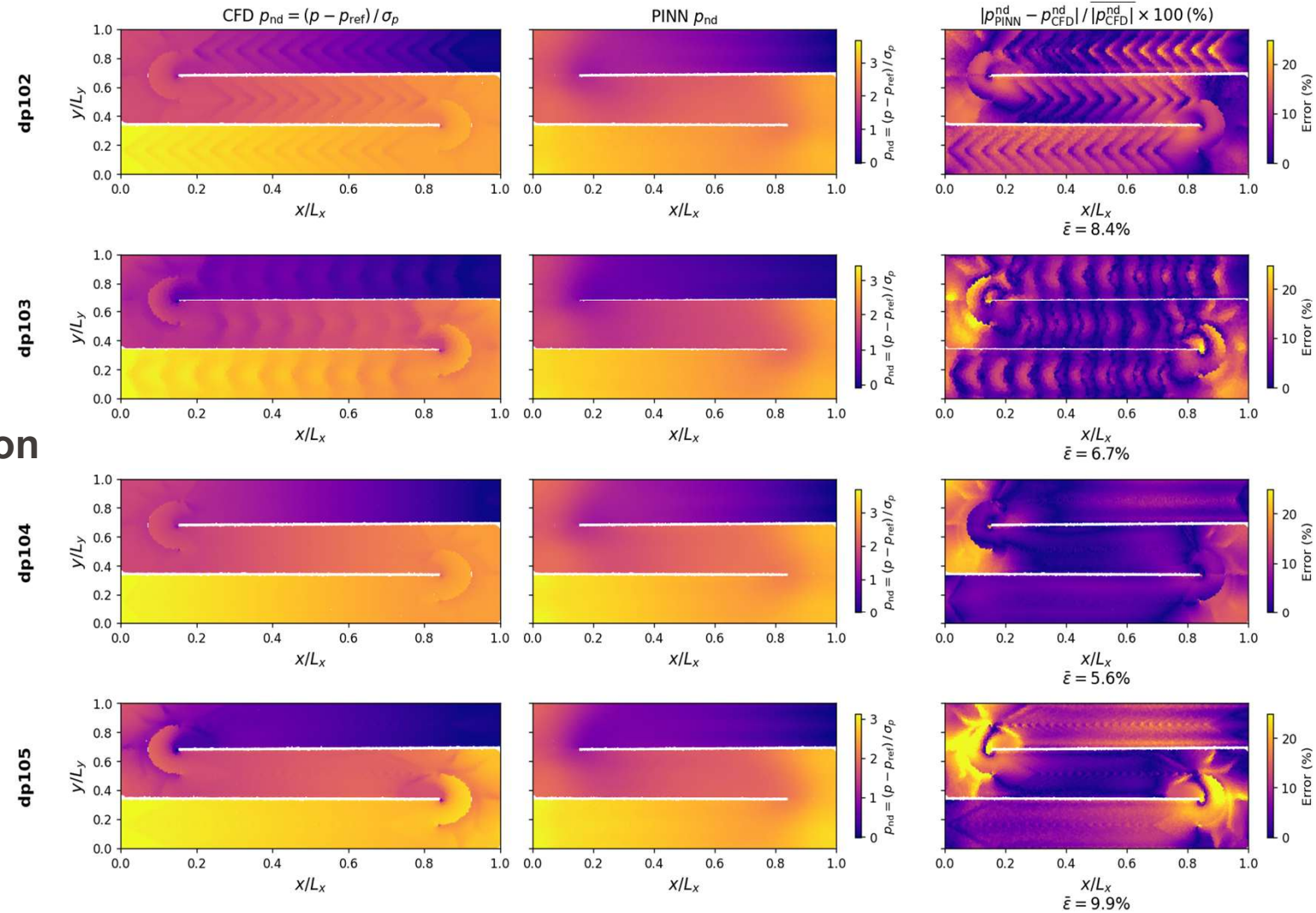


- Temperature field well captured ($\sim 11\%$ error)
- Errors concentrated near bends

- Global velocity distribution consistently captured (~25% error)
- Highest errors behind the ribs: recirculation zones create locally complex flow that is harder to resolve



Normalized Pressure: CFD, PINN pred, error on 4 unseen design points



II. MLP to predict heat flux:

- Heat Transfer Coefficient (HTC):

$$\text{HTC} = \frac{q''}{T_w - T_b}$$

From the MLP

From the PINN

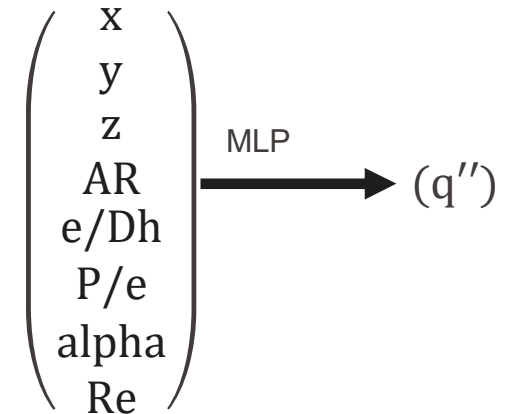
Differential form [\[edit\]](#)

The differential form of Fourier's law of thermal conduction shows that the local **heat flux** density $\mathbf{q}$ is equal to the product of **thermal conductivity** κ and the negative local temperature gradient $-\nabla T$. The heat flux density is the amount of energy that flows through a unit area per unit time.

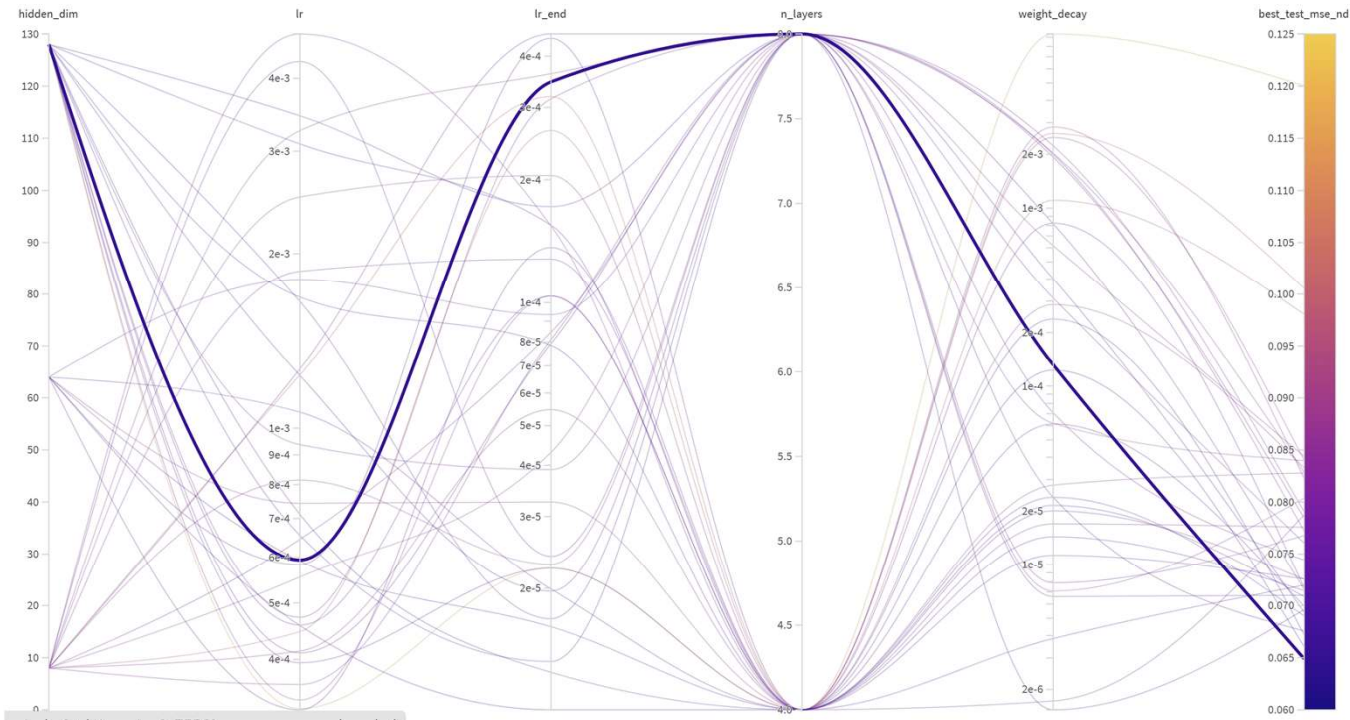
$$\mathbf{q} = -\kappa \nabla T,$$

where (including the **SI** units)

- $\mathbf{q}$ is the local heat flux density, W/m^2 ,
- κ is the material's **conductivity**, $\text{W}/(\text{m}\cdot\text{K})$,
- ∇T is the temperature gradient, K/m .



- **Target quantity:** local heat flux q''
- **First approach: Fourier's law:** requires temperature gradient in the boundary layer, where PINN resolution is insufficient
- **Fully data-driven MLP trained directly on CFD data input:** spatial coordinates (x, y, z) + 5 design parameters → output: q''



Chosen hyperparameters:

- hidden_dim: 128
- lr: 0.00059
- lr_end: 0.00034
- n_layers: 8
- Weight_decay: 0.00013

→ Training in 7 min

→ Inference in less than 1 second

Hyperparameter search: parallel coordinates plot and correlation table

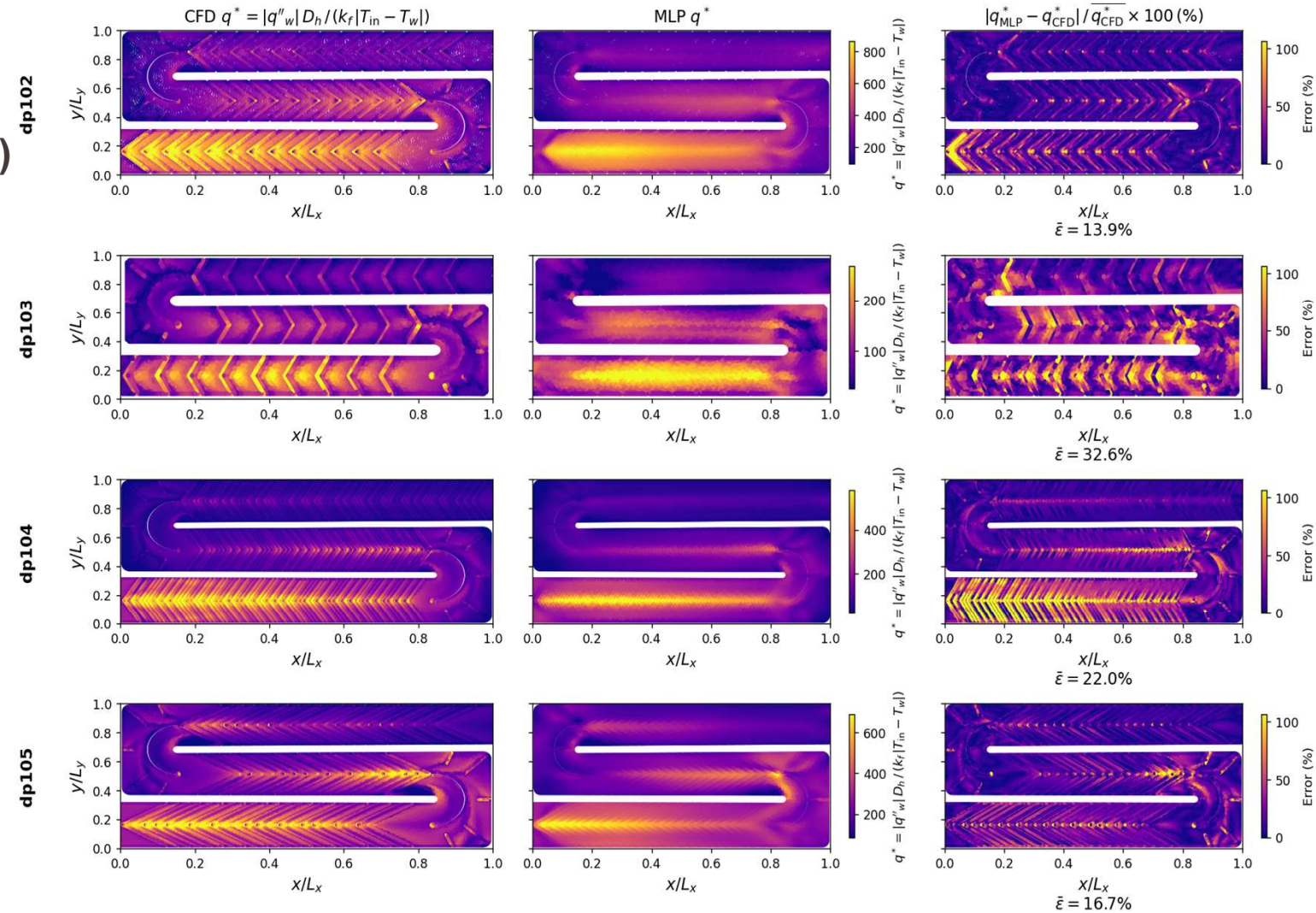
Config parameter	Importance ⓘ ↓	Correlation
weight_decay		
hidden_dim		
n_layers		
Runtime		
lr_end		
lr		

- Overall heat flux distribution well captured (13 - 32% error)
- Highly dependent on the design point

→ Model averages out the rib effect

→ MLP is more sensitive to position in the design space than the PINN (data driven learning)

Normalized heat flux: CFD, MLP pred, error on 4 unseen design points



III. Heat transfer coefficient:

- Heat Transfer Coefficient (HTC):

$$\text{HTC} = \frac{q''}{T_w - T_b}$$

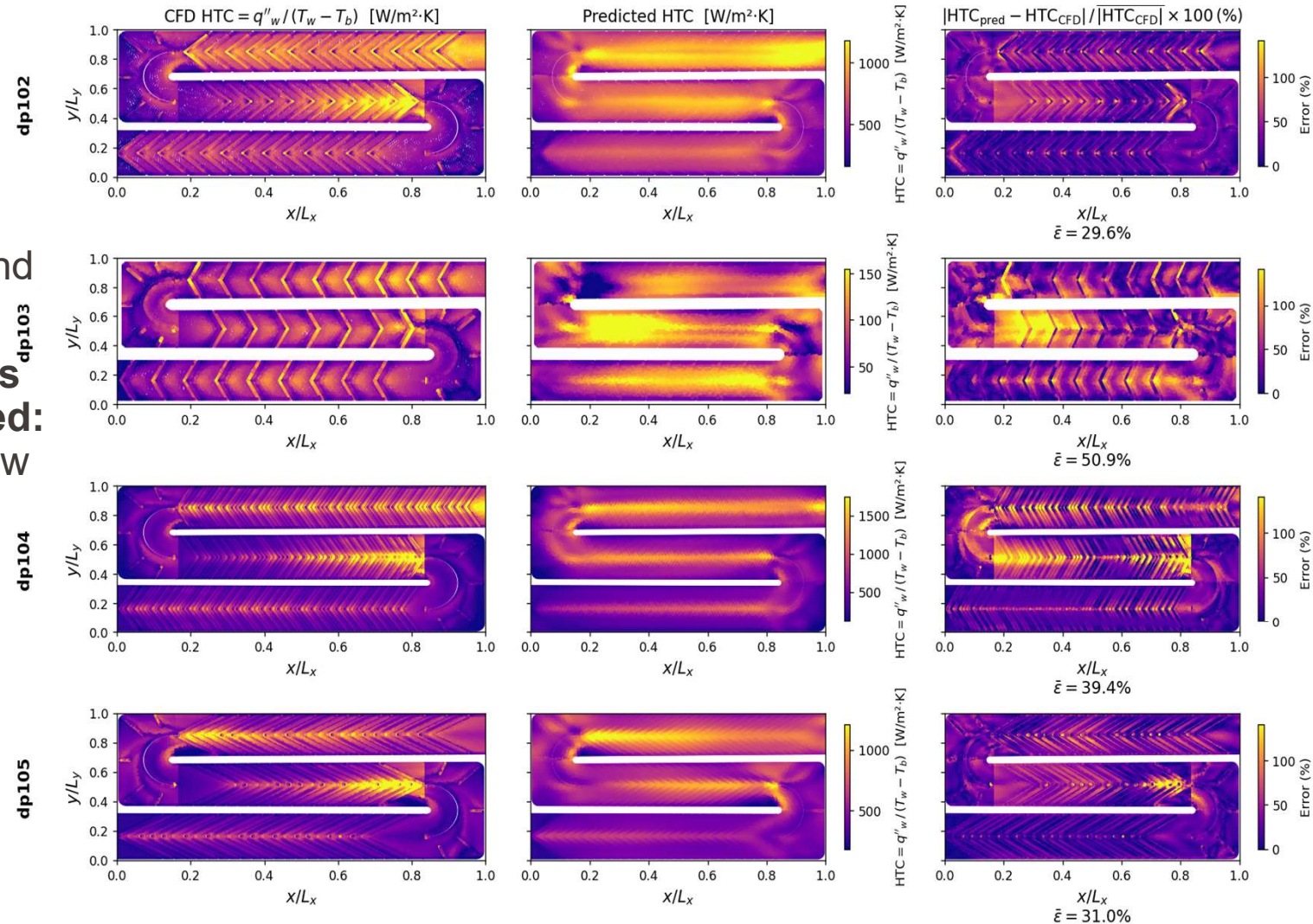
From the MLP

From the PINN

III. Heat transfer coefficient:

Heat transfer coefficient: CFD, PINN + MLP pred, error on 4 unseen design points

- **Errors up to 100% near rib:**
compound effect of the MLP and the PINN
- **Large-scale variations across the domain correctly captured:**
model distinguishes high vs. low heat transfer regions
- **Absolute local accuracy is imperfect, but inter-design-point variation is preserved**



IV. Conclusion:



IV. Conclusion :

- **PINN** successfully trained on 50 design points
 - Large-scale flow physics well recovered across all fields
 - local rib features remain hard to resolve for smooth networks
- **MLP surrogate** provides a practical alternative for heat flux where PINN resolution is insufficient
 - Design point dependent
- **Both models provide inter-design-point variation**

Future work:

- **Spectral bias:** neural networks preferentially learn **low-frequency** components of a function, struggling to resolve sharp, high-frequency features
- **Explore domain decomposition → Derive heat flux directly from the XPINN's predicted temperature field?**
- **Try optimization:** Integrate the surrogate into Anupam's optimization framework for efficient search over the five-parameter design space
- Try more advanced NS equation (not averaged)
- Try recurrent models

■

